

R-Squared and Adjusted R-Squared

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Introduction

R^2 is the proportion of variance in the outcome explained by the regression — the most commonly reported summary of regression fit. Its weakness is well known: R^2 never decreases when predictors are added, so it cannot distinguish a useful predictor from a noise variable, and high R^2 alone does not imply a useful model. Adjusted R^2 penalises for the number of predictors and decreases when an added predictor does not earn its complexity. For predictive purposes, cross-validated R^2 is more honest still.

Prerequisites

A working understanding of multiple regression, residual sum of squares, and the bias-variance trade-off in model selection.

Theory

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}.$$

It equals the squared correlation between observed and fitted values for OLS with an intercept. Always between 0 and 1; always non-decreasing as predictors are added.

Adjusted R^2 corrects for the penalty:

$$R^2_{\text{adj}} = 1 - (1 - R^2) \cdot \frac{n - 1}{n - p - 1},$$

with p the number of predictors. Adjusted R^2 can decrease when an added predictor does not improve fit enough to compensate for the lost degree of freedom; it can even be negative for very poorly fitting models.

For prediction, both are in-sample summaries; cross-validated R^2 ($1 - \text{SSE}_{\text{CV}} / \text{SS}_{\text{tot}}$) replaces the residual sum with a CV-based estimate of out-of-sample error and corrects for the optimism of in-sample fit.

Assumptions

Standard regression assumptions for the underlying linear model. R^2 is well-defined for any model with an intercept; without an intercept, definitions differ across software.

R Implementation

```
fit <- lm(mpg ~ wt + hp + disp, data = mtcars)
s <- summary(fit)
c(R2 = s$r.squared, R2_adj = s$adj.r.squared)

# Cross-validated R^2
library(caret)
set.seed(2026)
```

```
cv <- train(mpg ~ wt + hp + disp, data = mtcars,
            method = "lm",
            trControl = trainControl(method = "cv", number = 10))
cv$results
```

Output & Results

`summary(fit)$r.squared` and `$adj.r.squared` give in-sample values; `caret::train()` returns cross-validated R^2 and RMSE under 10-fold CV. The gap between in-sample and CV R^2 quantifies optimism — large gaps indicate overfitting.

Interpretation

A reporting sentence: “The model explained 83 % of the variance in mpg ($R^2 = 0.83$, adjusted $R^2 = 0.81$); 10-fold cross-validated R^2 was 0.79, indicating modest optimism in the in-sample estimate. Adjusted R^2 confirmed that all three predictors contributed beyond their penalty.” Always report adjusted R^2 alongside R^2 for multi-predictor models, and CV R^2 when prediction is the goal.

Practical Tips

- Report adjusted R^2 , not just R^2 , whenever predictors number more than one; the unadjusted version misleads about model parsimony.
- Adjusted R^2 can be negative for very poorly fitting models; this is informative, not a bug.
- For predictive modelling, prefer cross-validated R^2 or RMSE; in-sample R^2 is optimistic by design.
- Do not compare R^2 across different datasets with different outcome variance; the comparison is meaningless.
- Avoid causal language: “the model explains 83 % of the variance” is correct; “the model predicts 83 % of the outcome” is not.
- For GLMs, pseudo- R^2 measures (Cox-Snell, Nagelkerke, McFadden) generalise the concept; report the specific variant used because they give different numerical answers.

R Packages Used

Base R `summary.lm()` and `summary.glm()`; `performance::r2()` for tidy R^2 across model classes; `caret` and `tidymodels` for cross-validated R^2 ; `MuMIn::r.squaredGLMM()` for marginal and conditional R^2 in mixed models.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook’s distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)

- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI